

Figure 1. PRISMA-2020 Flow (Cumulative: Original Review + Update Window)

IDENTIFICATION

Records — ORIGINAL
(Jan 2018 - Jan 2026)
n = 462
(175 systematic + 287 AI)

Records — UPDATE
(29 Jan - 26 Apr 2026)
n = 1,192
(7 databases + snowball)

SCREENING

Records after dedup + PD-relevance filter: *n* = 334
(286 original + 48 update)

Title/abstract excluded: *n* = 236 (236 original; update used full-text triage)

ELIGIBILITY

Full-text articles assessed:
n = 98 **(50 original + 48 update)**

Full-text excluded: *n* = 38

ORIG (*n*=20): no validation 8, conference 5, diagnostic-only 4, duplicate 3

UPDATE (*n*=18): E1 leakage 9, E2 no-comp 3, E3 epi 2, E4 conference 2, Tier-C 2

INCLUDED

Studies INCLUDED in qualitative + quantitative synthesis
n = 60 **(30 original + 30 update)**

Pooled meta-analysis: 11 head-to-head studies (7 originals + 4 update)
Tier-2 external validation: 16 studies · PROBAST L:M:H = 24:56:17 (1%)
Mechanistic-DT (NASEM-partial): Hemedan 2026 (3/4) · Matsui 2026 (2/4)

Cumulative: 1,654 records → 334 after dedup → 98 full-text → 38 excluded → 60 included.